

NETWORKS AND COMPLEXITY

Solution 16-12

*This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 16.12: The complicated example revisited [Lvl. 4]

Let's return to the system from Ex. 16.3. Could you use the approach from the implicit function theorem also to compute $x''(p)$ at $p = 1$? How about $x'''(p)$ and $x''''(p)$? Use this to construct $x^*(p)$ as a Taylor expansion. Use your result to find $x^*(0)$, and verify that it is a steady state.

Solution

So we start with

$$f(x(p), p) = 0 \quad (1)$$

Differentiating this equation with respect to p gives us

$$f_p + f_x x' = 0, \quad (2)$$

as we saw in the chapter. So, let's differentiate again

$$f_{pp} + 2f_{xp}x' + f_{xx}(x')^2 + f_x x'' = 0 \quad (3)$$

and again

$$f_{ppp} + 3f_{ppx}x' + 3f_{xpp}(x')^2 + f_{xxx}(x')^3 + 2f_{xp}x'' + f_{xx}x'x'' + f_{xp}x'' + f_x x''' = 0 \quad (4)$$

and again

$$\begin{aligned} f_{pppp} + 4f_{pppx}x' + 3f_{xpp}x'' + 6f_{xxpp}(x')^2 + 6f_{xpp}x'x'' + 4f_{xxx}(x')^3 + f_{xxxx}(x')^4 \\ + 3f_{xxx}(x')^2x'' + 2f_{xpp}x'' + 2f_{xpp}x'x'' + 2f_{xp}x''' + f_x x'''' = 0 \end{aligned} \quad (5)$$

OK this was admittedly some tough differentiation, so let's do some simpler derivatives to relax a bit. In the example our function is

$$f = -x^4 + 2px^3 + 2p^2x - px^2 + x - 2p \quad (6)$$

Hence,

$$f_p = 2x^3 + 4px - x^2 - 2 = 16 + 8 - 4 - 2 = 18 \quad (7)$$

$$f_x = -4x^3 + 6px^2 + 2p^2 - 2px + 1 = -32 + 24 + 2 - 4 + 1 = -9 \quad (8)$$

$$f_{pp} = 4x = 8 \quad (9)$$

$$f_{px} = 6x^2 + 4p - 2x = 24 + 4 - 4 = 24 \quad (10)$$

$$f_{xx} = -12x^2 + 12px - 2p = -48 + 24 - 2 = -26 \quad (11)$$

$$f_{ppp} = 0 \quad (12)$$

$$f_{ppx} = 4 \quad (13)$$

$$f_{pxx} = 12x - 2 = 24 - 2 = 22 \quad (14)$$

$$f_{xxx} = -24x + 12p = -48 + 12 = -36 \quad (15)$$

$$f_{pppp} = 0 \quad (16)$$

$$f_{pppx} = 0 \quad (17)$$

$$f_{ppxx} = 0 \quad (18)$$

$$f_{pxxx} = 12 \quad (19)$$

$$f_{xxxx} = -24 \quad (20)$$

Now let's consider Eq. (2) again

$$0 = f_p + f_x x' \quad (21)$$

$$0 = 18 - 9x' \quad (22)$$

Therefore $x' = 2$ at $p = 1$, but we already knew this. Let's take a look at Eq. (3) instead

$$0 = f_{pp} + 2f_{xp}x' + f_{xx}(x')^2 + f_x x'' \quad (23)$$

$$0 = 8 + 2 \cdot 24 \cdot 2 - 26 \cdot 4 - 9x'' \quad (24)$$

$$0 = 8 + 4 \cdot 24 - 4 \cdot 26 - 9x'' \quad (25)$$

$$0 = -9x'' \quad (26)$$

Hence $x'' = 0$. Now in the next equation, Eq. (4) we can start by removing all the terms that contain a factor of x'' :

$$0 = f_{ppp} + 3f_{xpp}x' + 3f_{xpp}(x')^2 + f_{xxx}(x')^3 + 2f_{xp}x'' + f_{xx}x'x'' + f_{xp}x'' + f_x x''' \quad (27)$$

$$0 = f_{ppp} + 3f_{xpp}x' + 3f_{xpp}(x')^2 + f_{xxx}(x')^3 + f_x x''' \quad (28)$$

$$0 = 0 + 3 \cdot 4 \cdot 2 + 3 \cdot 22 \cdot 4 - 36 \cdot 8 - 9x''' \quad (29)$$

$$0 = 8 + 88 - 96 - 3x''' \quad (30)$$

$$0 = -3x''' \quad (31)$$

We are lucky, $x''' = 0$. Now there is only one left and this one will have many vanishing terms. I will start by putting in the $x''' = x'' = 0$ and then substitute the other terms:

$$0 = f_{pppp} + 4f_{pppx}x' + 3f_{xpp}x'' + 6f_{xxpp}(x')^2 + 6f_{xxp}x'x'' + 4f_{xxxp}(x')^3 + f_{xxxx}(x')^4 + 3f_{xxx}(x')^2x'' + 2f_{xpp}x'' + 2f_{xxp}x'x'' + 2f_{xp}x''' + f_x x'''' \quad (32)$$

$$0 = f_{pppp} + 4f_{pppx}x' + 6f_{xxpp}(x')^2 + 4f_{xxxp}(x')^3 + f_{xxxx}(x')^4 + f_x x'''' \quad (33)$$

$$0 = 4 \cdot 12 \cdot 8 - 24 \cdot 16 - 9x'''' \quad (34)$$

$$0 = -9x'''' \quad (35)$$

So $x'''' = 0$. We can see now that all subsequent terms will work out in the same way. either the terms are zero because they contain higher derivatives of f or they contain derivatives of x that we already know to be zero. We can now write the Taylor expansion

$$x(p) = x(1) + x'(1)(p-1) + \frac{1}{2}x''(1)(p-1)^2 + \dots \quad (36)$$

Since x'' and all higher derivatives are zero we are left with

$$x(p) = x(1) + x'(1)(p-1) \quad (37)$$

We know from the Ex. 16.3 that $x(1) = 2$ and computed here and previously that $x'(1) = 2$. there we have

$$x(p) = 2 + 2(p - 1) \tag{38}$$

So this branch of steady states is simply

$$x^* = 2p \tag{39}$$

Hence $x^* = 0$ should be a steady state for $p = 0$. We can confirm that this is indeed true by realizing that every term in the equation of motion contains either x or p as a factor.

Bonus: In the design of this exercise I cheated a bit to make sure that the result would be simple, and some nice cancellations would happen on the way. You can actually discover how I constructed the original system by solving the equation of motion for p . If you don't see it immediately use polynomial long division to remove the known solution $x = 2p$. It should then be quite simple to also compute the two bifurcation points in this system where a fold bifurcation and an unusual transcritical bifurcation occur.